# MCKV Institute of Engineering

**Project Title:** **Employee Performance Dashboard**

**Submitted by:** Sayantani Dalui

**Exam Roll No.:** 11620324008

**Department:** B.VOC in Software Development

**Semester:** 3rd Sem

A Power BI Project Report for Industrial Training at

**Ardent Computech Pvt. Ltd**

# Project Introduction

This project focuses on developing an Employee Performance Dashboard using Power BI to analyze workforce data and generate meaningful insights. It aims to evaluate employee performance, understand distribution across departments, and identify key factors affecting attrition. The dashboard provides interactive visualizations that support data-driven decision-making and help improve overall organizational efficiency.

### Data Visualization

HR datasets transformed into actionable workforce insights

### Interactive Dashboard

Dynamic visuals with slicers and cross-filtering

### Strategic Decision-Making

Data-driven HR decisions supported by visual analytics

# Project Objectives

The objective of this project is to analyze employee data using Power BI to evaluate performance and workforce distribution.
It also aims to identify key factors influencing attrition and support data-driven decision-making.

01

## Workforce Overview

To understand how employees are distributed across departments and locations, along with gender distribution and count of total Employee.

02

## Performance Analysis

To assess employee productivity based on key performance metrics like Projects handled, working hours etc.

03

## Attrition Insights

To identify patterns and factors contributing to employee turn-over, salary correlations, overtime effects, and tenure patterns

04

## Strategic Planning

Data-driven decisions for retention, training programs, and workforce optimization.

# Key Dashboard Features

### Interactive Visualizations

Bar charts, donut charts, line charts, maps, and matrix tables for comprehensive data exploration

### Dynamic Slicers

Age group, department, job role, work-life balance, and gender filters for targeted analysis

### Theme Toggle

Dark mode and light mode switch for improved visual comfort and accessibility

### Multi-Page Layout

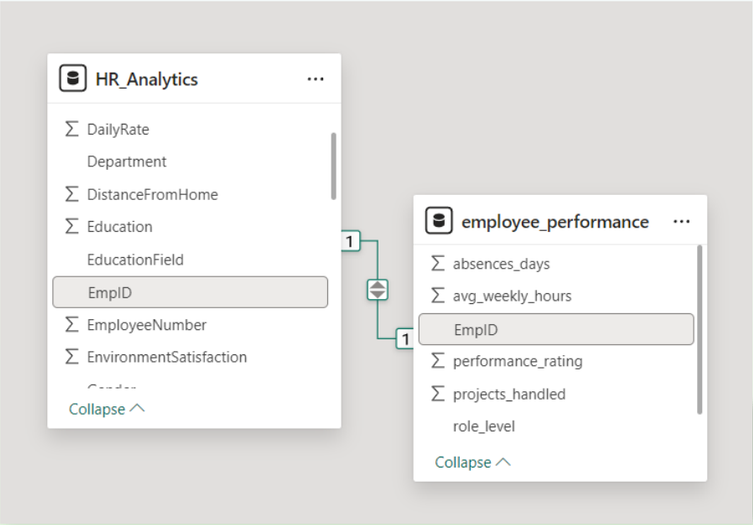
Employee overview, performance metrics, and attrition analysis across dedicated report pages

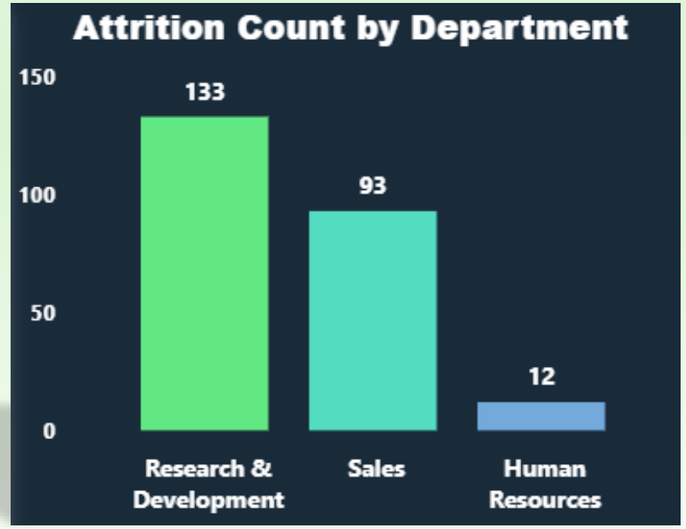
### KPI Indicators

Card visuals displaying total employees, average income, years at company, and attrition rate

### Cross-Filtering

Interactive filtering between visuals to analyze relationships across different variables




# Implementation Methodology

### Data Collection

HR Analytics and Employee Performance datasets from Kaggle with 1,480 employee records

### Data Cleaning

Power Query Editor used to remove duplicates, verify data types, and ensure consistency

### Data Modeling

Two tables connected via EmpID with one-to-one relationship for accurate analysis

### Visualization

DAX measures and Visualization for KPIs, charts, and interactive dashboard elements

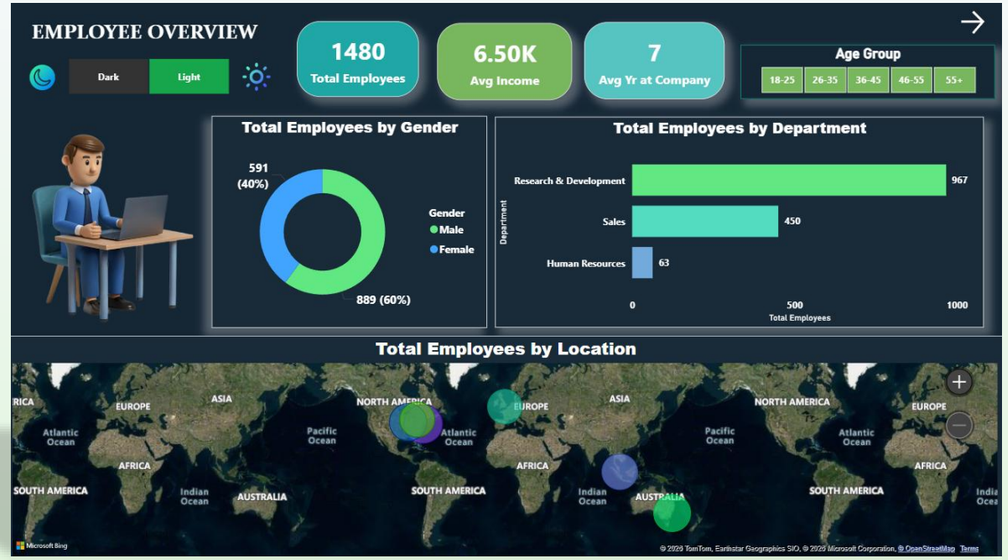
# Key Insights from Analysis

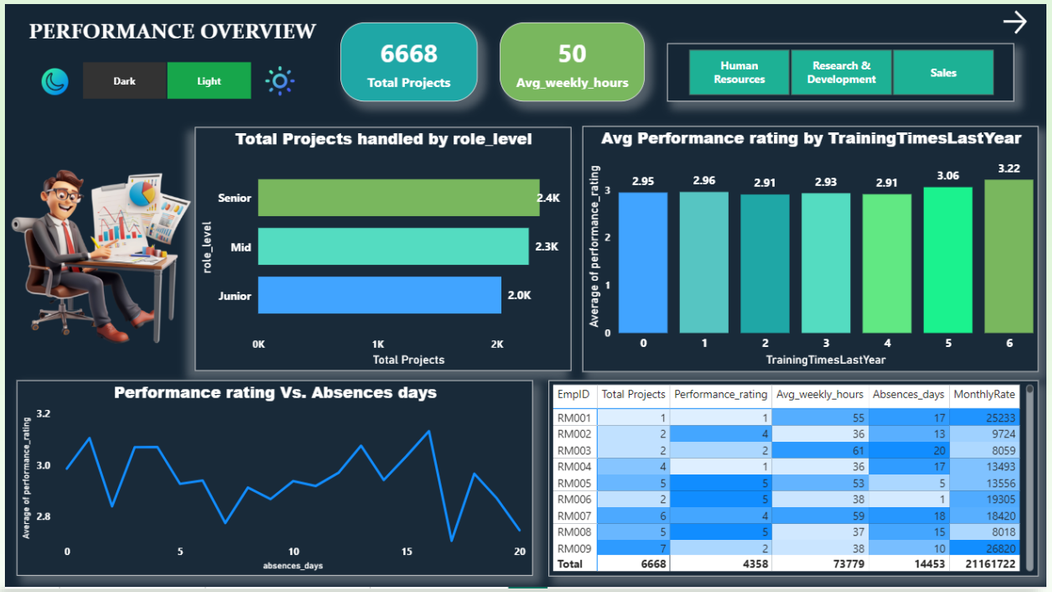
## Workforce Profile

- 1,480 total employees with 60% male and 40% female distribution
- Research & Development department has highest employee count
- Average tenure of 7 years indicates workforce stability
- Average monthly income of $6,500 across organization
- The Map chart reveals a geographically diverse workforce.

## Performance Trends

- 6,668 total projects handled reflecting productivity levels
- Average 50 weekly working hours indicating workload
- Senior employees contribute more significantly to projects
- Training shows positive relationship with performance ratings
- Absenteeism alone is not a reliable indicator of performance.




| EmpID | Total Projects | Performance_rating | Avg_weekly_hours | Absences_days | MonthlyRate |
|---|---|---|---|---|---|
| RM001 | 1 | 1 | 55 | 17 | 25233 |
| RM002 | 2 | 4 | 36 | 13 | 9724 |
| RM003 | 2 | 2 | 61 | 20 | 8059 |
| RM004 | 4 | 1 | 36 | 17 | 13493 |
| RM005 | 5 | 5 | 53 | 5 | 13556 |
| RM006 | 2 | 5 | 38 | 1 | 19305 |
| RM007 | 6 | 4 | 59 | 18 | 18420 |
| RM008 | 5 | 5 | 37 | 15 | 8018 |
| RM009 | 7 | 2 | 38 | 10 | 26820 |
| Total | 6668 | 4358 | 73779 | 14453 | 21161722 |

# Attrition Analysis Findings

**238**

**Employees Left**

Total number who departed organization

**16.08%**

**Attrition Rate**

Overall turnover percentage

**Early Years**

**High Risk Period**

Most attrition occurs during initial employment years

**Overtime**

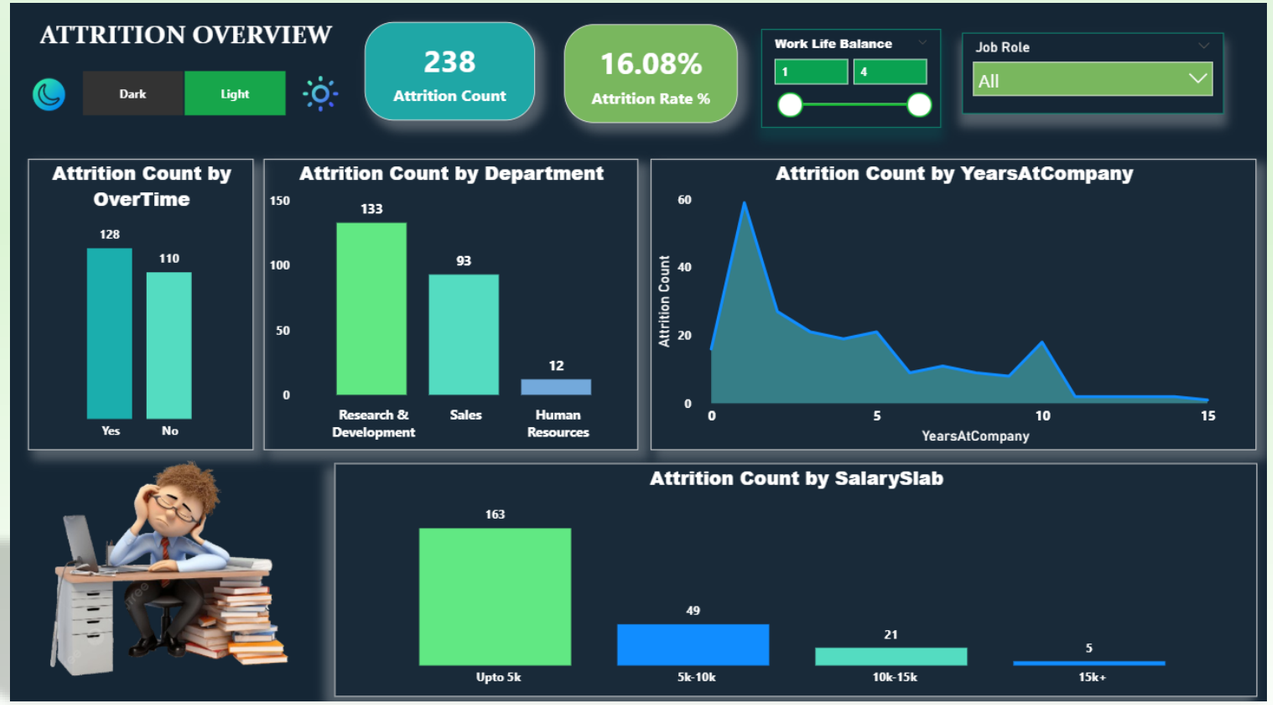
**Key Factor**

Employees working overtime more likely to leave

## Others

Lower salary groups show higher attrition compared to higher salary groups.

Research & Development department experiences highest turnover.



# Advantages and Limitations

## Advantages

- Interactive and user-friendly dashboard for easy data analysis.
- Dynamic filtering using slicers for better data exploration.
- Comprehensive view of employee performance, distribution, and attrition in a single dashboard.
- Efficient use of Power BI features like DAX, visuals, and data modeling.
- Flexible design that can be adapted to other domains also.

## Limitations

- Dataset is static (Kaggle), so no real-time data updates
- Analysis is limited to the available fields in the dataset
- No predictive analytics or advanced machine learning techniques applied.
- Desktop-based dashboard limits accessibility and sharing.
- Requires Power BI license for full functionality.
- Data quality and accuracy depend on the original dataset, which may affect the reliability of insights.

# Conclusion and Future Scope

## Conclusion

The Employee Performance Dashboard successfully demonstrates how data analytics transforms raw employee data into actionable workforce insights. Through interactive visualizations, organizations can monitor performance, identify attrition patterns, and support data-driven HR decisions.

## Future Enhancements

**Real-Time Integration**
Connect to live HR databases for monitoring metrics in real time

**Predictive Analytics**
Machine learning models to predict employee attrition risk

**Cloud Deployment**
Power BI Service for secure access and collaboration across departments

**Mobile Optimization**
Responsive design for monitoring insights on mobile devices

# Thank You

Special thanks to MCKV Institute of Engineering & Ardent Computech Pvt. Ltd., for guidance and support throughout this training project. Thank you so much for listening.

~ By Sayantani Dalui